Final exam:
Mathematics for Artificial Intelligence 2

Duration: 180 minutes
Maximum of 50 points

Rules:

- **Put your name and matriculation nr. on every sheet!**
- Use a new sheet for every exercise!
- Electronic devices are not allowed!
- Do not use pencil or a red pen.
- The whole procedure of solution is required.
- All results must be simplified as far as possible.

Good luck!

EXERCISE 1: **Derivatives** (4 points)

Find the derivatives of the following functions:

- $f_1(x) = \dfrac{x}{2 + \sin(x^2)}, \qquad x \in \mathbb{R}$
- $f_2(x) = \ln\left(3 \cdot 2^x\right), \qquad x \in \mathbb{R}$

Simplify the results, and determine the value of the derivative at $x_0 = \sqrt{\pi}$.

EXERCISE 2: **Limits** (4 points)

Determine if the following proper or improper limits exist and, if applicable, determine the value.

$$\text{a)} \quad \lim_{x \to 0} \frac{\cos(x) - 1}{x^2} \qquad\qquad \text{b)} \quad \lim_{x \to 1} \frac{x^3 - 2x + 1}{(x+1)^2}$$

EXERCISE 3: **Global and Local Extrema** (10 points)

Consider the function $f\colon [0, \infty) \to \mathbb{R}$ with $f(x) = x \cdot \mathrm{e}^{-x}$.

a) Determine all local and global extrema and extreme values of f.

b) Where is f monotonically increasing/decreasing?

c) Where is f convex/concave?

EXERCISE 4: **Taylor series** (10 points)

Determine the Taylor polynomial $T_3(x)$ of order 3 and the Taylor series $T(x)$ of the function

$$f\colon \mathbb{R} \to \mathbb{R},\ f(x) = (x^2 - 3)^2$$

around $x_0 = 1$. Where does the Taylor series converge?

EXERCISE 5: **Integrals** (6 points)

Compute $\displaystyle\int_0^{\pi} x^2 \cdot \sin(x)\,\mathrm{d}x$.

EXERCISE 6: **Fourier series** (8 points)

Compute all Fourier coefficients and the Fourier series $Sh(x)$ of the function $h\colon [0,1) \to \mathbb{R}$ with $h(x) = e^{\pi i x}$. Simplify the result! (Complex numbers in canonical form.)
Does the Fourier series converge in $(0,1)$? Justify your answer!
(Ignore the point $x = 0$.)

EXERCISE 7: **Multivariate derivatives** (8 points)

Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x,y) := \begin{cases} \dfrac{xy}{x-y}, & \text{if } x \neq y, \\ 0, & \text{if } x = y \end{cases}$$

Where is f partially/totally differentiable? Find the gradient there.